Regency Centers — Trade Area Purchasing Power

An Interactive Predictive Analytics Dashboard | CEN6940.14925 | Group 7 | Spring 2026

445 trade areas | **5** divisions | **3** ML models

Section 0 Project Background & Context

🏢 About This Project

Regency Centers is a leading U.S. owner, operator, and developer of grocery-anchored shopping centers. The company currently evaluates trade areas using its proprietary "DNA" model — a descriptive framework based on household income, population, and growth rates. While informative, this model is static and lacks predictive capability.

This project replaces descriptive analysis with a full predictive analytics pipeline: Exploratory Data Analysis → PCA-based Purchasing Power Index construction → Machine Learning modelling → Interactive data storytelling.

Analytical Framework (Rogers et al., 2004): *Purchasing Power = Income Capacity × Market Size*

📊 Dataset Overview

- 530 original trade area records from the Regency Centers demographic dataset
- **445 valid trade areas** retained after cleaning (removing missing values)
- 8 demographic & economic variables (2025 projections)
- 5 geographic divisions: West (90), Northeast (95), Southeast (136), Central (74), Mid-Atlantic (50)
- Historical data available from 2017 through 2025

❓ Four Research Questions

1. Which demographic and economic variables most predictively drive trade area purchasing power?
2. What is a consistent, scalable approach to measuring purchasing power across the portfolio?
3. What growth and decline trends characterize purchasing power across trade areas?
4. How accurately and reliably can purchasing power be predicted from demographic variables?

📋 Eight Variables Used in Analysis

INCOME CAPACITY

- Median Household Income (2025)
- 5-Year Income Growth Projection
- Average Home Value (2025)
- Education Attainment (% Bachelor's+)

MARKET SIZE

- Residential Population (2025)
- Daytime Population (2025)
- Total Households (2025)
- 5-Year Household Growth

Filter by Division

All Divisions

Filter by Income Range ($39k – $267k)

$39k $93k $112k $138k $267k

TRADE AREAS	AVG INCOME	AVG POPULATION	AVG PPI SCORE
445	$117,801	124,055	-0.00

TOTAL PP ($B)	AVG INCOME GROWTH	AVG HH GROWTH
$6,212.1B	14.6%	2.7%

Section 1 Market Overview — Portfolio Purchasing Power

💡 **Key Finding: West and Mid-Atlantic divisions deliver the highest average purchasing power per trade area ($18.3B and $18.9B). Southeast has the most properties (136) but the lowest average ($8.5B) — reflecting broad suburban coverage rather than density concentration.**

Figure 1: Top 10 Markets by Purchasing Power ($B)

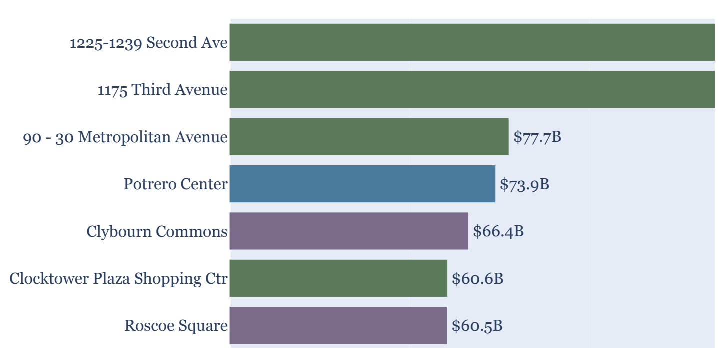
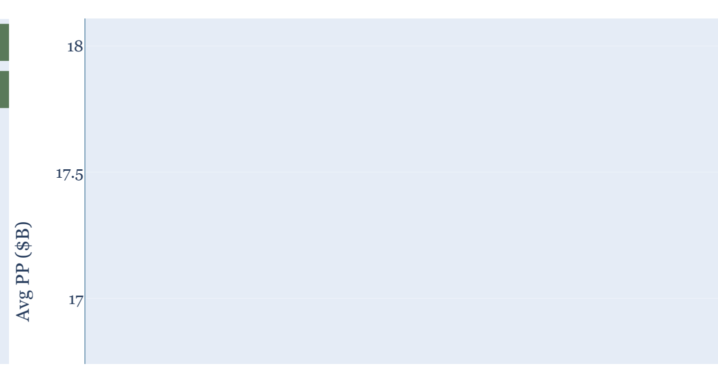


Figure 2: Average Purchasing Power by Division



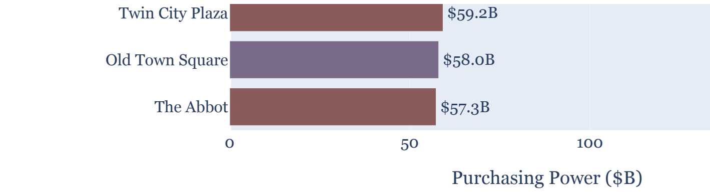


Figure 3: Average PPI Score by Division

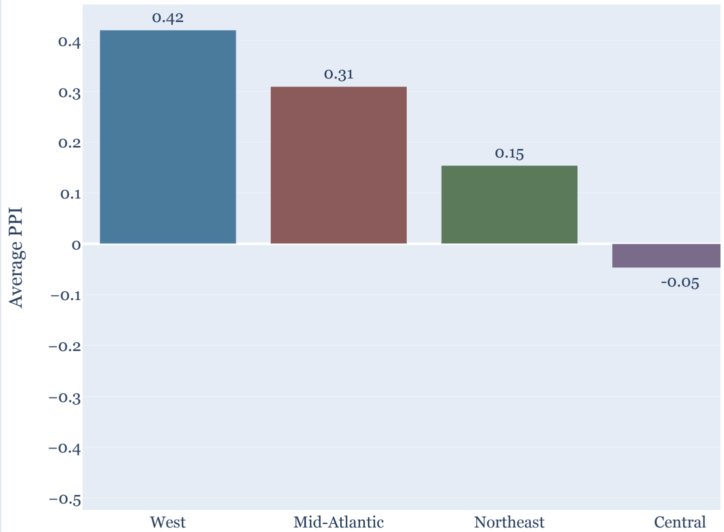
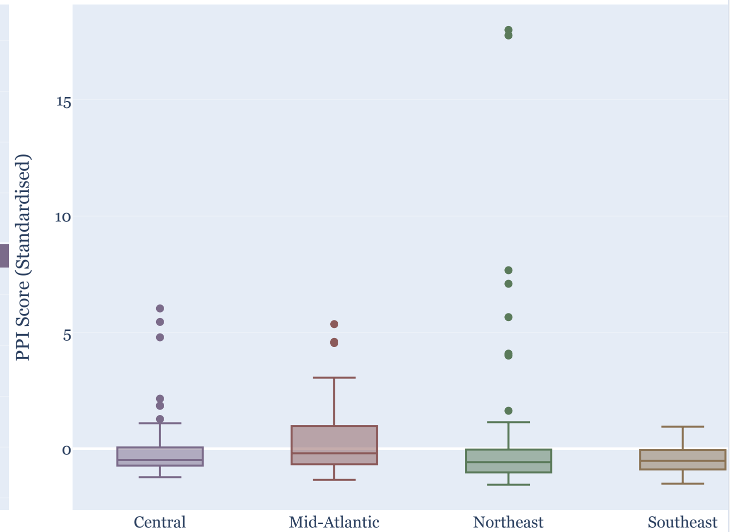


Figure 4: PPI Distribution by Division



What These Charts Show

Top 10 Markets (upper left): The two New York City properties — 1225–1239 Second Ave ($167.5B) and 1175 Third Avenue ($165.2B) — dwarf all other markets. Their combined purchasing power ($332.7B) exceeds that of the entire Southeast division ($1.16T across 136 properties on a per-property basis). This extreme concentration requires a distinct investment and tenant strategy.

Division Box Plot (lower right): Northeast shows the widest distribution with the highest outliers (PPI = 18), while West and Mid-Atlantic show tight distributions consistently above the portfolio mean — reliable, not outlier-driven performance. Southeast is narrow and below the mean.

Section 2

Demographic Drivers — What Predicts Purchasing Power?

💡 **Critical Finding: Population (r = 0.937) and Daytime Population (r = 0.928) are overwhelmingly stronger predictors of purchasing power than Median Income (r = 0.108). Market volume — not income per household — drives aggregate retail spending capacity.**

Figure 1a: Population vs. Purchasing Power

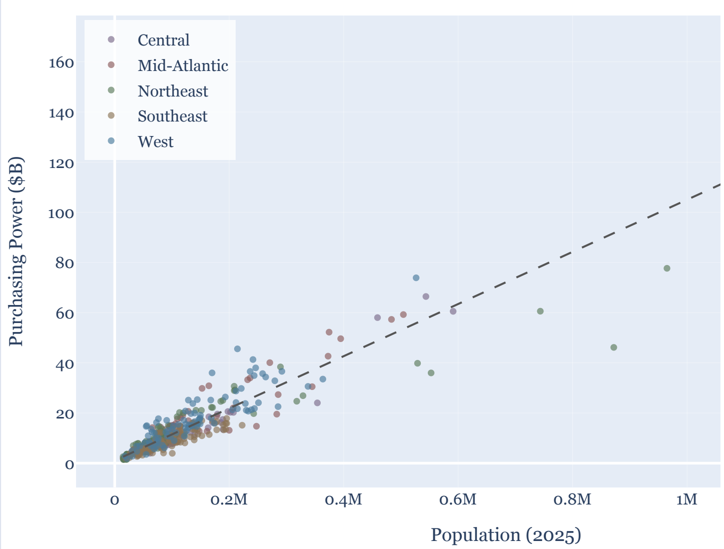


Figure 1b: Median Income vs. Purchasing Power

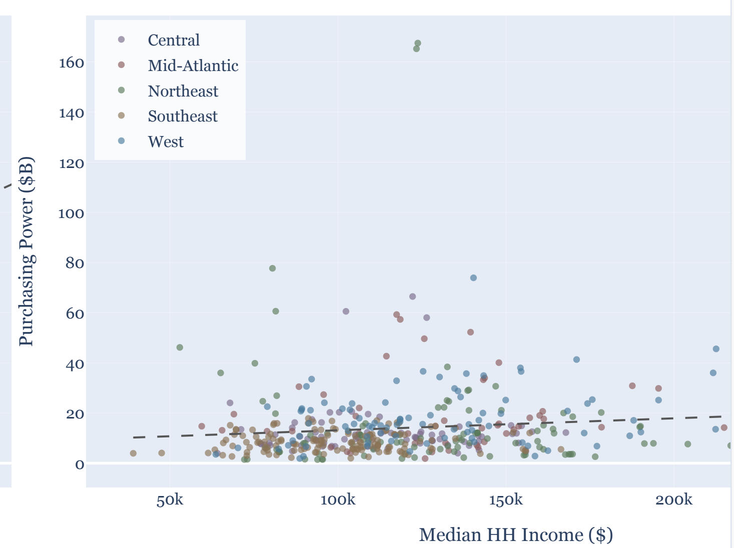
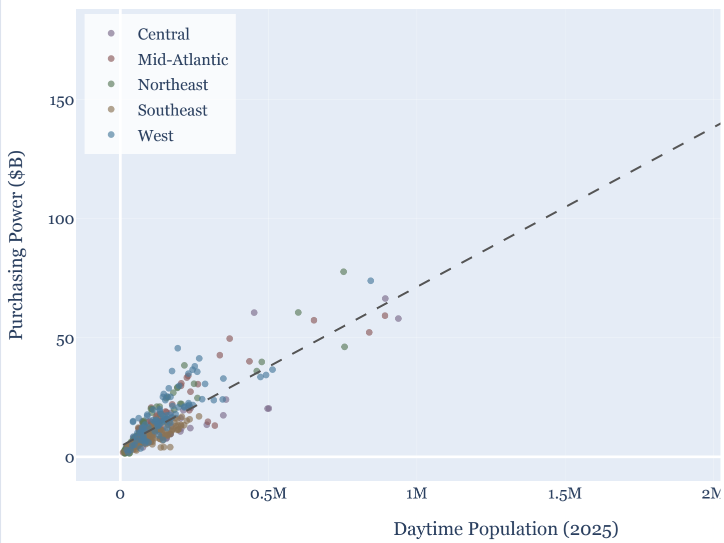
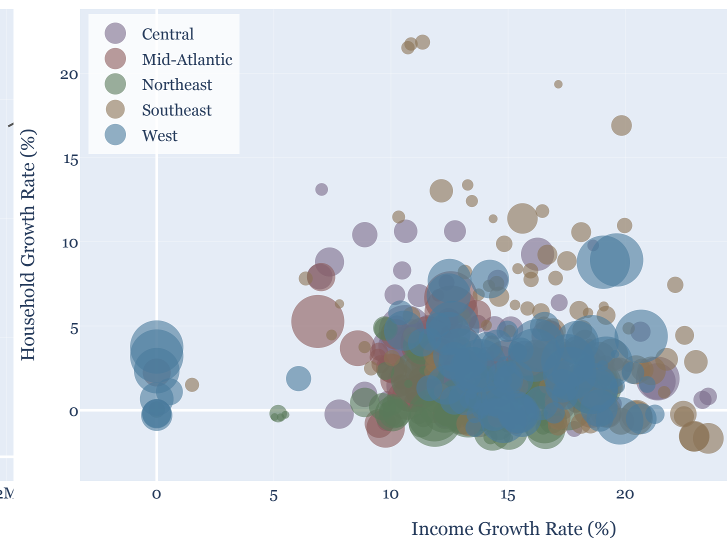


Figure 1c: Daytime Population vs. Purchasing Power

Figure 1d: Income Growth vs. HH Growth (bubble = Purchasing Power)





🔍 How to Read These Charts

Population vs. PP (upper left): A steep, tight relationship — the larger the residential market, the greater the aggregate spending capacity. West and Northeast trade areas dominate the upper right; Southeast clusters in the lower left.

Income vs. PP (upper right): The relationship is nearly flat. High-income trade areas scatter widely across all purchasing power levels. A $267k-income area with 20,000 residents generates far less total spending than a $90k-income area with 500,000 residents.

Growth Bubble (lower right): Markets in the upper-right quadrant — above-average income AND household growth simultaneously — are emerging strategic opportunities. Small bubbles in this quadrant have modest current purchasing power but strong dual-growth momentum, making them ideal for early investment ahead of demand growth.

Figure 5: Correlation Matrix — All Demographic Variables vs. PPI

	Income	Income_Growth	Home_Value	Education	Population	Daytime_Pop	Households	HH_Growth	PPI
Income	1	−0.57	0.6	0.63	−0.14	−0.08	−0.13	−0.13	−0.05
Income_Growth	−0.57	1	−0.22	−0.3	0.11	0.09	0.11	−0.02	0.1
Home_Value	0.6	−0.22	1	0.47	0.22	0.23	0.22	−0.09	0.34
Education	0.63	−0.3	0.47	1	−0.01	0.14	0.08	0.06	0.15
Population	−0.14	0.11	0.22	−0.01	1	0.91	0.98	−0.06	0.97
Daytime_Pop	−0.08	0.09	0.23	0.14	0.91	1	0.96	0.02	0.97
Households	−0.13	0.11	0.22	0.08	0.98	0.96	1	−0.02	0.99
HH_Growth	−0.13	−0.02	−0.09	0.06	−0.06	0.02	−0.02	1	−0.04
PPI	−0.05	0.1	0.34	0.15	0.97	0.97	0.99	−0.04	1

Section 3

Principal Component Analysis — The Purchasing Power Index (PPI)

💡 **PCA Finding: PC1 (the PPI) explains 37.6% of total portfolio variance and loads primarily on market size variables. The near-zero loading of Median Income (−0.027) confirms that income per household and market volume are nearly orthogonal — independent strategic dimensions.**

🧮 Why PCA? Why the PPI?

The conventional Income × Population measure combines only two variables. PCA constructs the Purchasing Power Index (PPI) by simultaneously weighting all eight demographic variables according to their natural covariance structure — producing a composite index that is data-driven rather than assumed.

The PPI correlates with Income × Population at r = 0.96 (confirming both capture the same phenomenon), but is methodologically superior because it:

① Incorporates all eight variables simultaneously

② Resolves multicollinearity by construction through orthogonal components

③ Reveals the two-dimensional portfolio structure (PC1 = market size; PC2 = income quality)

Figure 6: PCA Scree Plot — Variance Explained by Component

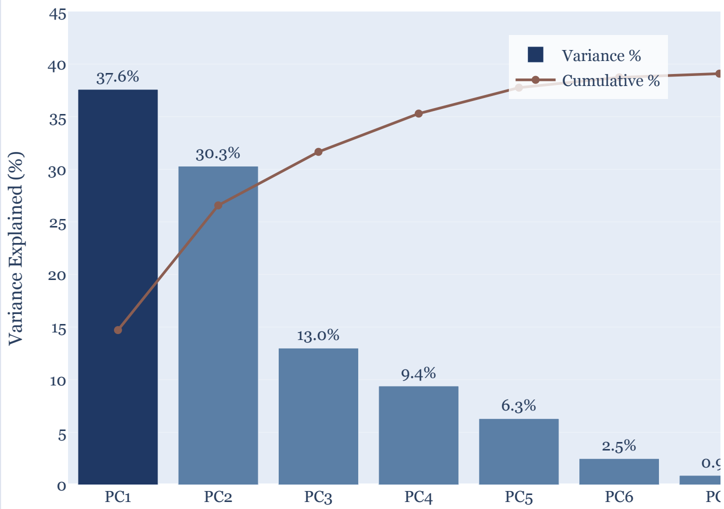


Figure 7: PC1 Loadings — What Drives the Purchasing Power Index

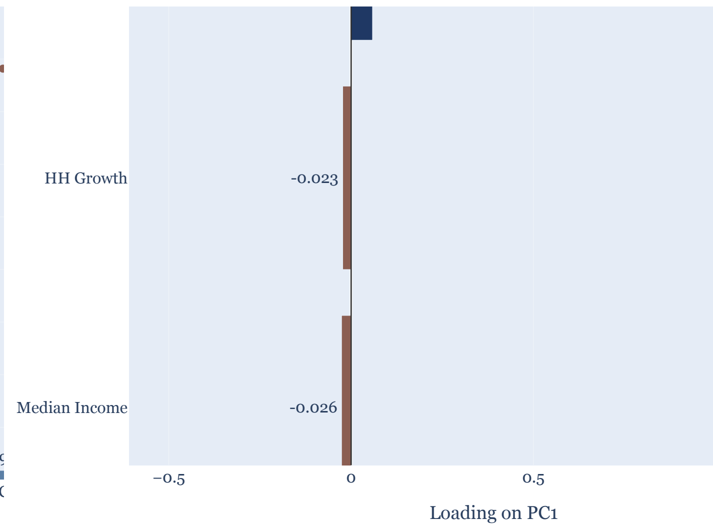


Figure 2a: Population vs. Purchasing Power Index (PPI)

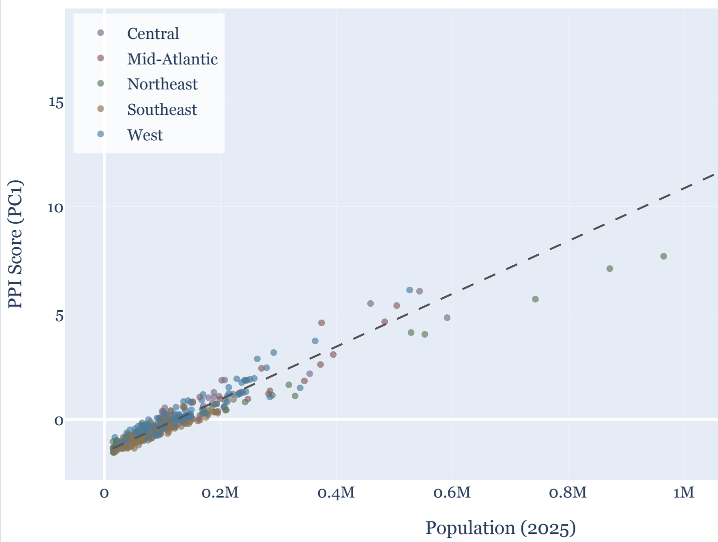
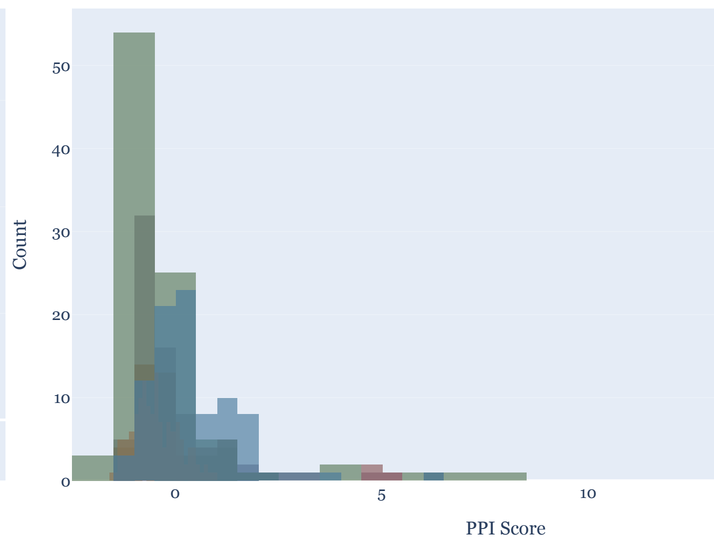


Figure 2b: PPI Score Distribution by Division



The Two-Dimensional Portfolio Strategy Framework

Because PC1 (market size) and PC2 (income quality) are orthogonal, every trade area can be positioned in a two-dimensional space. This reveals four strategic quadrants:

① Premium Urban Core

High PC1 + High PC2

Flagship tenants, premium mix, highest investment priority

② Volume Market

High PC1 + Low PC2

Grocery anchors, value retail, high-traffic convenience

③ Affluent Suburban

Low PC1 + High PC2

Specialty retail, experiential, boutique service

④ Standard Suburban

Low PC1 + Low PC2

Core grocery anchor, operational efficiency focus

Section 4

Predictive Models — Machine Learning Results

💡 **Model Result: All three models achieve test R^2 above 0.91 on genuinely held-out data. Gradient Boosting (R^2 = 0.936, MAE = 0.163 PPI units) is recommended for operational trade area scoring. Linear Regression (CV R^2 = 0.947) is recommended for stakeholder communication due to its interpretable coefficients.**

🤖 Why Three Models?

Linear Regression
Provides interpretable coefficients that explain why a specific trade area scores as it does. Best cross-validation R² (0.947) confirms stable generalisation. Essential for stakeholder communication.

Random Forest
200 decision trees on bootstrapped samples. Captures non-linear interactions without parametric assumptions. Directly addresses the normality and homoscedasticity violations found in regression assumption testing.

Gradient Boosting ✓ Recommended
200 sequential trees — each corrects the residual errors of the previous ensemble. Achieves the lowest test MAE (0.163 PPI units) and highest test R². Recommended for operational trade area scoring.

Figure 8: Model Performance Comparison — Test Set + Cross-Validation

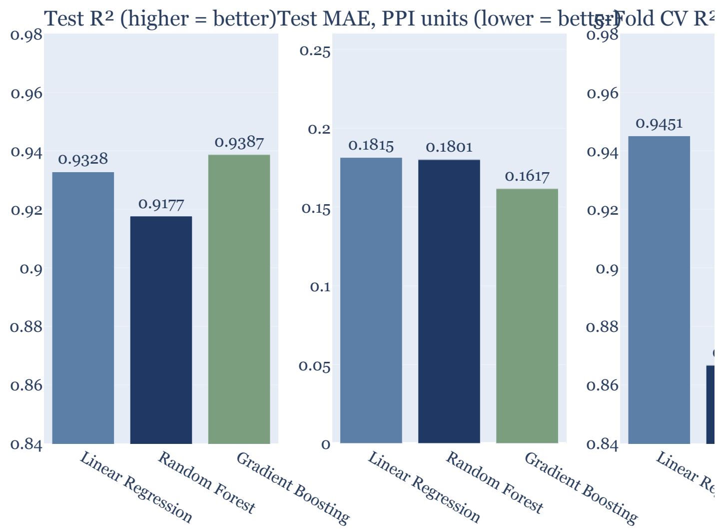
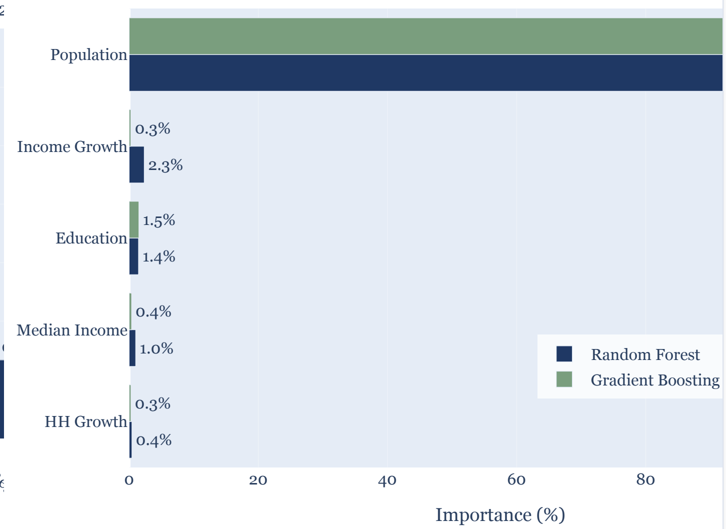
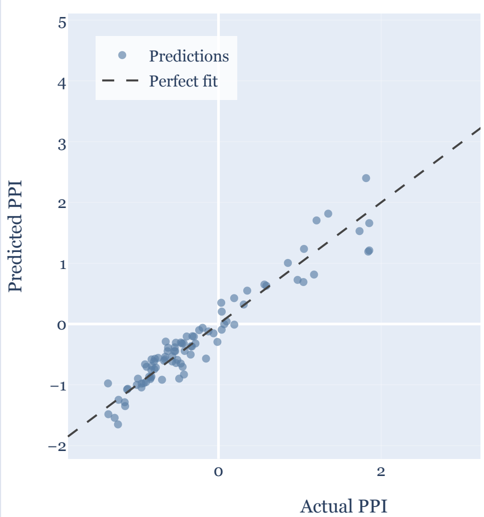


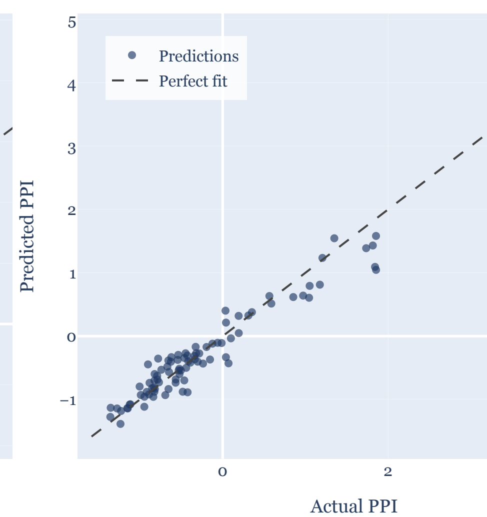
Figure 9: Feature Importance — Random Forest vs. Gradient Boosting



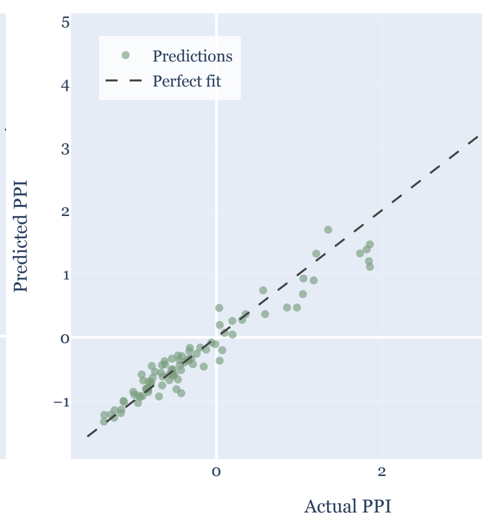
Linear Regression — Actual vs. Predicted
R² = 0.933 MAE = 0.1815



Random Forest — Actual vs. Predicted
R² = 0.918 MAE = 0.1801



Gradient Boosting — Actual vs. Predicted
R² = 0.939 MAE = 0.1617



⚖️ How to Read the Feature Importance Chart

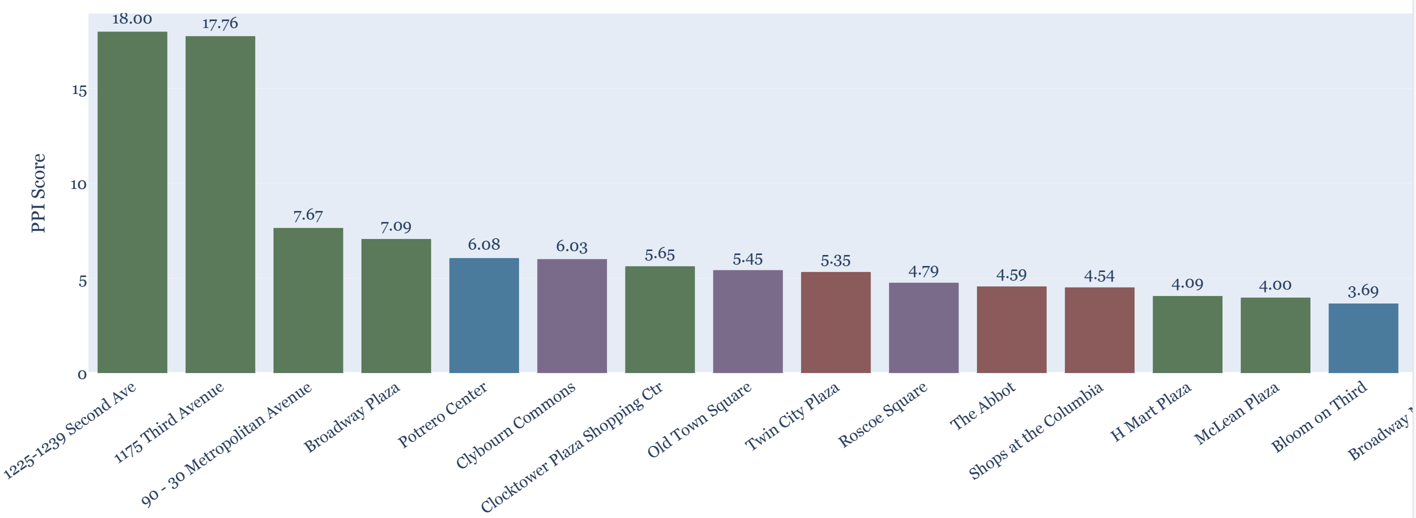
Population accounts for 93.8% (Random Forest) and 97.8% (Gradient Boosting) of all predictive importance. This is consistent with correlation analysis (r = 0.937), PCA (PC1 loading +0.559), and regression (t = 81.7, p < .001). Every analytical method converges on the same conclusion: population volume is the primary driver of trade area purchasing power.

Income contributes less than 1% of feature importance across both models — consistent with its near-zero PC1 loading (−0.027). This directly challenges its role as a co-equal site-selection criterion in the existing DNA model. Income is better used as a post-selection tenant-strategy input, not a primary investment filter.

Section 5 **Strategic Priorities — Top Trade Areas by PPI**

💡 **Actionable Output: Use the Division and Income filters above to instantly generate a ranked priority list for any market segment. This chart updates dynamically — applying any strategic lens without code changes.**

Figure 10: Top 20 Trade Areas by Purchasing Power Index (PPI)



Section 6 **Key Conclusions & Strategic Recommendations**

Conclusion 1 — Market Size is the Primary Driver

Population volume is the overwhelmingly dominant determinant of trade area purchasing power, confirmed by four independent analytical methods: correlation analysis ($r = 0.937$), PCA (PC1 loading +0.559), regression ($t = 81.7$, $p < .001$), and ML feature importance (93.8%–97.8%). Median Income contributes less than 1% of predictive importance.

Recommendation: Establish population density as the mandatory first-stage site-selection filter. Tier 1: above ~140,000 (75th percentile). Tier 4: below ~60,000 (25th percentile).

Conclusion 2 — Daytime Population Matters Equally

Daytime population has a PC1 loading of +0.559 — equal to residential population. For grocery-anchored retail, commuter and workforce traffic is often the primary revenue driver, especially near employment centers, transit hubs, and medical campuses.

Recommendation: Add daytime population to the site assessment scorecard with equal weight to residential population. This requires no new data infrastructure.

Conclusion 3 — Income Belongs in Tenant Strategy

Median Income ($r = 0.108$; PC1 loading −0.027; <1% ML importance) adds noise rather than signal to site selection. It operates on an independent dimension (PC2) that measures spending quality, not spending volume.

Recommendation: Remove income from site-selection scoring. Use it exclusively as a post-selection tenant-strategy input: high income →

Conclusion 4 — Deploy the Recommended Models

Gradient Boosting

For operational scoring of new trade areas

$R^2 = 0.936$ | MAE = 0.163 PPI units

Linear Regression

For stakeholder communication & investment rationale

CV $R^2 = 0.947$ | Interpretable coefficients

The dual-model approach correctly separates predictive accuracy (Gradient Boosting) from interpretability (Linear Regression) — both required for practical deployment in a commercial real estate investment context.

Conclusion 5 — Build an Emerging Market Pipeline

Markets with above-average income growth AND above-average household growth simultaneously — concentrated in Central and West divisions — represent the highest-priority forward-looking opportunities. Their current PPI scores may be modest, but dual-growth dynamics position them for higher purchasing capacity within 3–5 years.

Recommendation: Establish a formal quarterly review process tracking PPI trajectory, population growth realisation, and occupancy performance for these emerging candidates.

⚠️ Limitations & Future Work

- **Circular definition:** Income × Population shares variables with model predictors — log-transformation of PPI recommended
- **NYC metro outliers:** Two properties (PPI = 18.01, 17.76) cause normality violations and dominate portfolio metrics
- **Cross-sectional data:** Panel modeling using 2017–2024 historical data would detect PPI trajectories over time
- **Missing context:** Competitive retail density, foot traffic data, and physical accessibility data would enable capturable demand estimation

premium/specialty retail; high population + moderate income → grocery anchors and value convenience.

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